Type MS · Compare light and electron microscopes and use their parts

Answer 1 of 6 · Question 3 part c · 2 marks

2023 JUN · Paper 02 (regional R) · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) | • contains particular features / characteristics/ adaptations;<br>• to perform a specific function; | | 2 |
| (b) (i) | • 84 ÷ 0.006;<br>• 14000x | Allow correct final answer from a calculation that includes 84 and 0.006, max 1 mark | 2 |
| (ii) | half the amount/haploid/23 chromosomes; | | 1 |
| (iii) | • to provide energy;<br>• for swimming/movement/travelling; | | 2 |
| (iv) | • 180 ÷ 0.070;<br>• 2571;<br>• (÷ 60) 42.85; | Allow any figure that rounds to 43<br>Allow ecf for third mark - an incorrect value for a calculation carried out for mps1 and 2, divided by 60 **and** a correct answer to that calculation | 3 exp |
| (c) | Any two from:<br>• greater magnification;<br>• higher resolution/can see objects more clearly/in more detail;<br>• can see a greater number of structures; | | 2 exp |

**Total Question 3 = 12 marks**

Type MS · Compare light and electron microscopes and use their parts

Answer 2 of 6 · Question 2 part a.ii · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 2 (a) | (i) | <table><tr><th>Part</th><th>Name of part</th><th>Function</th></tr><tr><td>X</td><td>mitochondria;</td><td></td></tr><tr><td>Y</td><td></td><td>controls activities of the cell/stores DNA / genetic information;</td></tr></table> | | 1<br>1 |
| | (ii) | **A,** more cell structures can be seen with greater resolution | | 1 |
| | (iii) | 60mm/6cm;<br>60 ÷ 0.05;<br>1200; | Ecf two marks for correct calculation from incorrect measurement<br>Full marks for correct final answer | 3 |
| (b) | | • diffusion;<br>• from a high concentration (in the cell) to a lower concentration (in the blood)/down a concentration gradient; | | 1<br>1 |

**Total question 2 = 8 marks**

## Type MS · Compare light and electron microscopes and use their parts

### Answer 2 of 6 · Question 2 part a.ii · 1 mark

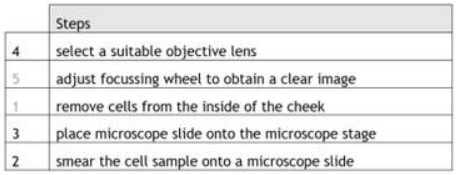
2019 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Compare light and electron microscopes and use their parts

Answer 3 of 6 · Question 3 part a.ii · 1 mark

2020 NOV · Paper 01 (regional R) · find part a.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 | (a) (i) |  | One mark for 4 before 5.<br>One mark for 3 before 2 | 2 |
| | (ii) | more detail/organelles can be seen/organelles/named organelle seen more clearly (1) | | 1 |
| | (iii) | reference no stain applied to specimen(1) | | 1 |
| | (iv) | • reference to sterilising (named) equipment (1)<br>• wash hands to avoid cross contamination(1)<br>• incubate/store in sealed petri dish / container | allow valid sterilisation technique | 2 |
| | (b) (i) | 10 × 40 = 400 | Full marks for correct answer without working | 1 |
| | (ii) | reference to 1000 (1)<br>0.018 × 1000 = 18 (µm) | Full marks for correct final answer | 2 |

**Total Question 3 = 9 marks**

## Type MS · Compare light and electron microscopes and use their parts

### Answer 4 of 6 · Question 5 part a.ii · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

| Question number | | | | | |
|---|---|---|---|---|---|
| 5 | (a) | (i) | • join with ovum;<br>• during fertilisation; | | 2 |
| | | (ii) | any two from<br>• ribosomes;<br>• endoplasmic reticulum/ER;<br>• chromosomes; | | 2 |
| | (b) | | • site of (aerobic) respiration;<br>• energy released/ATP produced;<br>• sperm motile/active/swim; | | 3 |
| | (c) | | 3µm = 0.003 mm;<br>distance between X and Y = 145 mm;<br>145;<br>0.003<br>=48333/48000; | ECF = 2 | 4 |

Total 11 marks

## Type MS · Compare light and electron microscopes and use their parts

### Answer 4 of 6 · Question 5 part a.ii · 2 marks

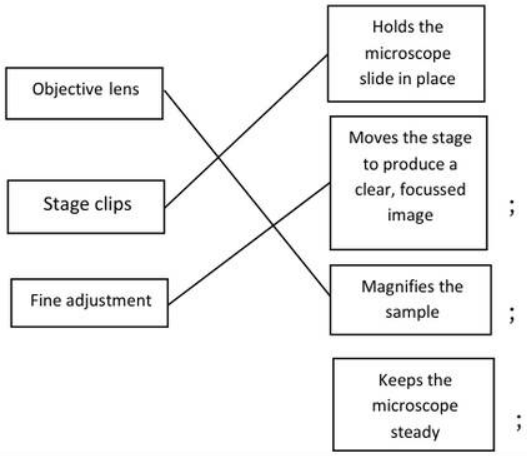
2021 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Compare light and electron microscopes and use their parts

Answer 5 of 6 · Question 1 part b.i · 3 marks

2024 JUN · Paper 01 (regional R) · find part b.i in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 1 (a) (i) | **Any two from:**<br>• (both) have a cell membrane;<br>• (both) have cytoplasm;<br>• (both) have genetic material/DNA;<br>• (both) contain ribosomes; | | (2) |
| (a) (ii) | **Any two from:**<br>• bacterial cell has a cell wall / an animal doesn't;<br>• bacterial cell has no nucleus /an animal cell does;<br>• bacterial cell has a plasmid /the animal cell doesn't;<br>• bacterial cell has a flagellum/ the animal cell doesn't;<br>• bacterial cell has pili /the animal cell doesn't;<br>• bacterial cell has no mitochondria /the animal cell does;<br>• bacterial cell has slime capsule / the animal cell doesn't; | | (2) |
| (b) (i) |  ; ; ; | Reject more than one line from each part | (3) |
| (b) (ii) | • substitution, 5 ÷ 50 000 = 0.0001(cm);<br>• conversion to mm - 0.0001 x 10;<br>• 0.001 mm/$1\times10^{-3}$;<br><br>or<br>• 5×10=50mm;<br>• 50÷50000;<br>• 0.001mm/$1\times10^{-3}$ | ecf for mp 2 and 3<br><br>Full marks for correct final answer | (3) |

**Total for Question 1 = 10 marks**

Type MS · Compare light and electron microscopes and use their parts

Answer 6 of 6 · Question 7 part c · 2 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

7 (a)

| Part | Name of part | Function of part |
|---|---|---|
| X | cell membrane; | controls what enters/leaves cell; |
| Y | nucleus; | controls cell/stores genetic information; |
| Z | cytoplasm; | site of chemical reactions; |

If they get any wrong in the part column, they can score marks for the correct function.

6

(b)

- scrape inside of cheek/cotton bud to remove cheek cells/eq.;
- place (scraping) on slide;
- smear/spread thinly;
- add stain/ methylene blue;
- place coverslip on top;
- remove excess stain;
- sterile/disinfect cotton bud;

4

(

(c)

- ribosomes;
- mitochondria;

accept any suitable structures such as ER,golgi,RER

ignore chromosomes

2

Total = 12

PAPER TOTAL - 90 MARKS

## Type MS · Compare light and electron microscopes and use their parts

### Answer 6 of 6 · Question 7 part c · 2 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

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